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Pertemuan 3 MA4171
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Fungsi Transfer dan Diagram Blok

MA4171 Teori Kontrol Linear

Latihan:

Tentukanlah fungsi transfer dari sistem berikut.

$$1. y'' + 3y' + 7y = 6u$$

$$2. y''' + 6y'' + 2y' + 4y = -5u'' + 8u'$$

$$3. \begin{cases} x_1' = -3x_1 + x_2 - 4u \\ x_2' = -2x_1 - u \\ y = x_1 - 2x_2 \end{cases}$$

$$4. y'' + 3y' + 7y = 6u_1 + 5u_2 - 4u_2'$$

$$5. \begin{cases} y_1'' + 6y_1' + 2y_1 = u' - 3u \\ y_2' + 6y_2 = 4u' \end{cases}$$

$$6. \begin{cases} y_1''' + 7y_1'' + 6y_1' + y_1 = u_1'' + 3u_1 + u_2 \\ y_2''' + 7y_2'' + 6y_2' + y_2 = 4u_2' \end{cases}$$

$$7. \begin{cases} y_1' + 2y_1 = u_1 + 5u_2 \\ y_1 + y_2' + 3y_2 = 4u_2 + u_2' \end{cases}$$

Representasi Sistem Dinamik

Menuliskan representasi system dinamik melalui fungsi transfer dan diagram blok

Model dalam bentuk persamaan ruang keadaan

- Model dalam bentuk time-variant:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

$$y(t) = C(t)x(t) + D(t)u(t)$$

- Model dalam bentuk time-invariant:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

- x : variabel keadaan (state variable), A : matriks keadaan (state matrix),
 u : kontrol masukan (input control), B : matriks masukan (input matrix),
 y : luaran (output), C : matriks luaran (output matrix).

- Model dalam bentuk time-invariant:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

- $Y(s) = G(s)U(s)$
- $G(s) = C(sI - A)^{-1}B + D$

Contoh:

- Sistem massa pegas dengan model $my''(t) + by'(t) + ky(t) = u(t)$.
Dengan mendefinisikan $y = x_1$, dan $y' = x_2$, persamaan ruang keadaannya dapat ditulis dalam bentuk:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u, \quad y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Contoh

$$\rightarrow \dot{\vec{x}} = A\vec{x} + Bu, \vec{x} \in \mathbb{R}^3$$

(1) Tentukan model linear dari persamaan $\ddot{y} + 3\ddot{y} + 2\dot{y} = u$ $A \in \mathbb{R}^{3 \times 3}$

(2) Tentukan fungsi transfer dari system berikut:

$$\dot{x}_1 = -4x_1 - x_2 + u$$

$$\dot{x}_2 = 3x_1 - x_2 + u$$

$$y = x_1$$

(1) Misal, $x_1 = y$, $x_2 = \dot{y}$ dan $x_3 = \ddot{y}$

diperoleh, $\dot{x}_1 = \dot{y} = x_2$

$$\dot{x}_2 = \ddot{y} = x_3$$

$$\begin{aligned} \dot{x}_3 = \ddot{y} &= -3\ddot{y} - 2\dot{y} + u \\ &= -2x_2 - 3x_3 + u \end{aligned}$$

Pers. Ruang Keadaan :

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = x_1 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} s+4 & 1 \\ -3 & s+1 \end{bmatrix}^{-1} = \frac{1}{(s+4)(s+1)+3} \begin{bmatrix} s+1 & -1 \\ 3 & s+4 \end{bmatrix} = \frac{1}{s^2+5s+7} \begin{bmatrix} s+1 & -1 \\ 3 & s+4 \end{bmatrix}$$

$$\text{Jadi, } \frac{Y(s)}{U(s)} = \begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \frac{1}{s^2+5s+7} \begin{bmatrix} s+1 & -1 \\ 3 & s+4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{s}{s^2+5s+7}$$

$$= \frac{1}{s^2+5s+7} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s \\ s+7 \end{bmatrix}$$

Perhatikan diagram blok: $u \rightarrow \boxed{G_1} \xrightarrow{v} \boxed{G_2} \xrightarrow{y}$

$$G_1: \dot{\vec{X}}_1 = A_1 \vec{X}_1 + B_1 u$$

$$v = C_1 \vec{X}_1 + D_1 u$$

G_1 : input u , output v
 G_2 : input v , output y

$$G_2: \dot{\vec{X}}_2 = A_2 \vec{X}_2 + B_2 v$$

$$y = C_2 \vec{X}_2 + D_2 v$$

$$\text{Maka } \dot{\vec{X}}_2 = A_2 \vec{X}_2 + B_2 (C_1 \vec{X}_1 + D_1 u)$$

$$y = C_2 \vec{X}_2 + D_2 (C_1 \vec{X}_1 + D_1 u)$$

$$\dot{\vec{X}}_1 = A_1 \vec{X}_1 + B_1 u$$

$$\dot{\vec{X}}_2 = B_2 C_1 \vec{X}_1 + A_2 \vec{X}_2 + B_2 D_1 u$$

$$y = D_2 C_1 \vec{X}_1 + C_2 \vec{X}_2 + D_2 D_1 u$$

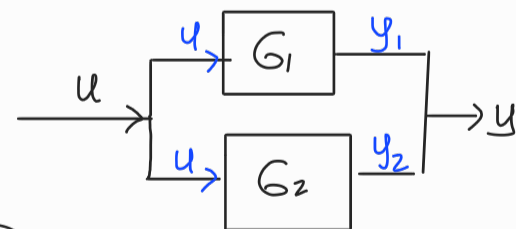
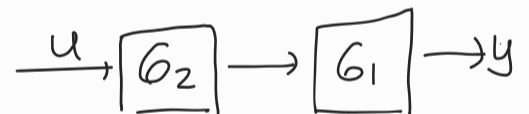
$$\begin{bmatrix} \dot{\vec{X}}_1 \\ \dot{\vec{X}}_2 \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ B_2 C_1 & A_2 \end{bmatrix} \begin{bmatrix} \vec{X}_1 \\ \vec{X}_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 D_1 \end{bmatrix} u$$

$$y = \begin{bmatrix} D_2 C_1 & C_2 \end{bmatrix} \begin{bmatrix} \vec{X}_1 \\ \vec{X}_2 \end{bmatrix} + D_2 D_1 u$$

$$\text{Fungsi Transfer: } \left[\begin{array}{cc|c} A_1 & 0 & B_1 \\ B_2 C_1 & A_2 & B_2 D_1 \\ \hline D_2 C_1 & C_2 & D_2 D_1 \end{array} \right]$$

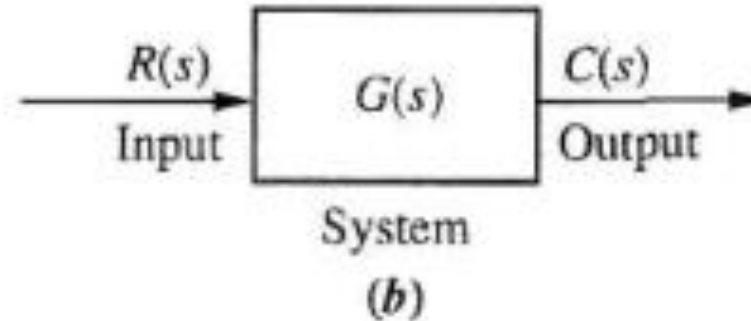
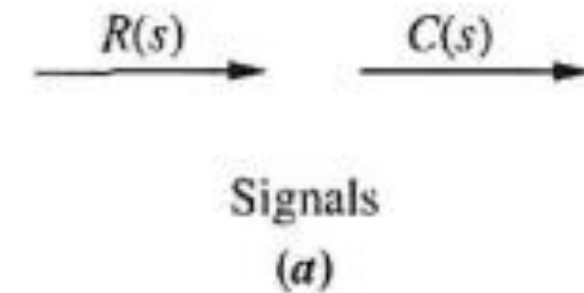
Latihan

Cari fungsi transfer

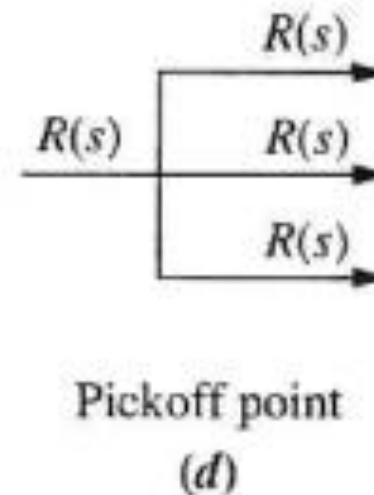
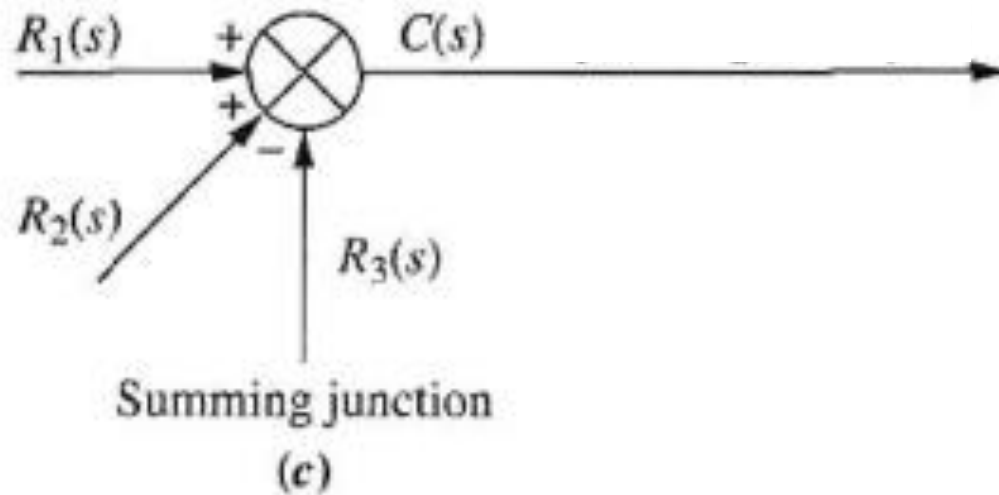


$$y = y_1 + y_2$$

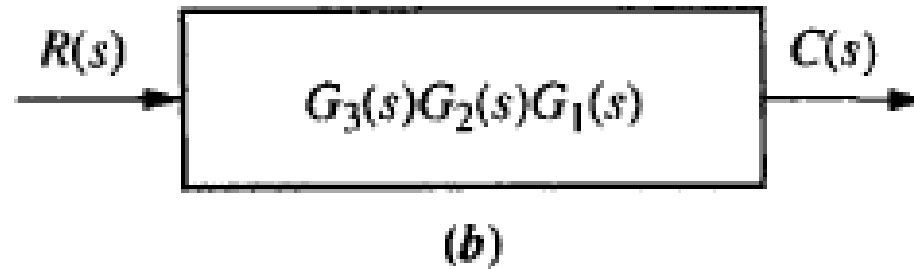
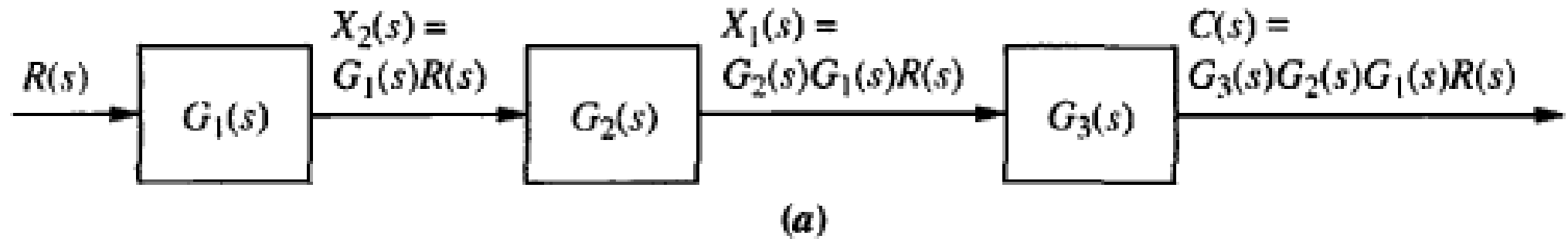
Diagram Blok: menggambarkan aliran sinyal



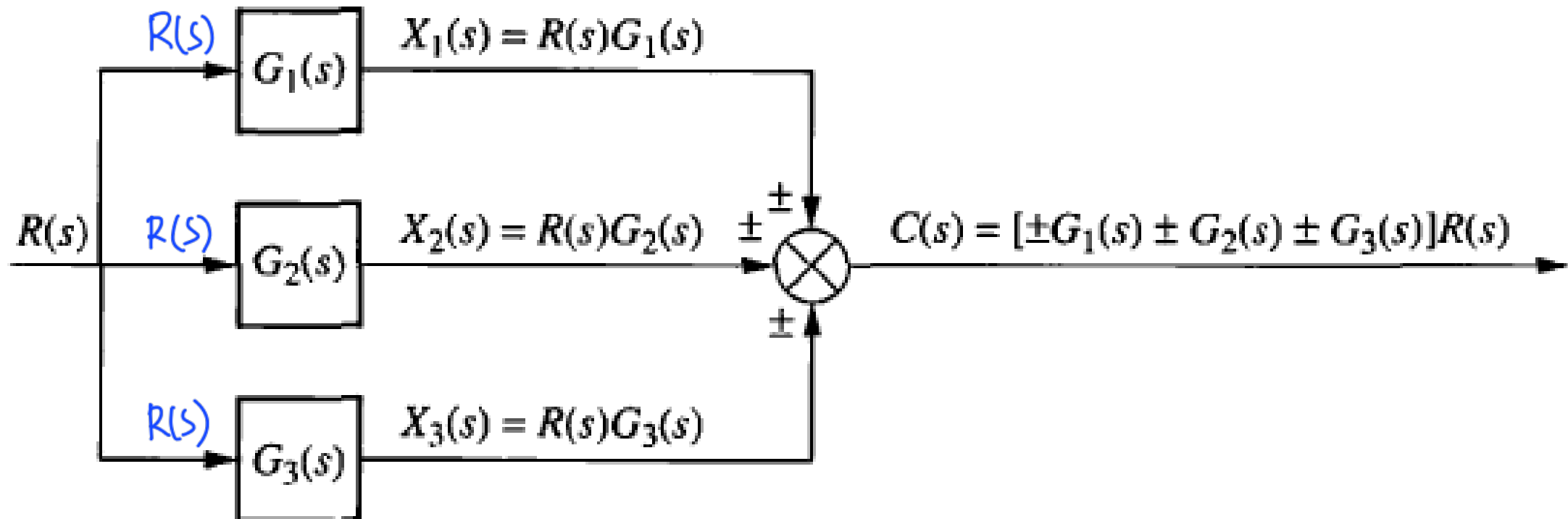
$$G(s) = \frac{C(s)}{R(s)}$$
$$C(s) = G(s) R(s)$$



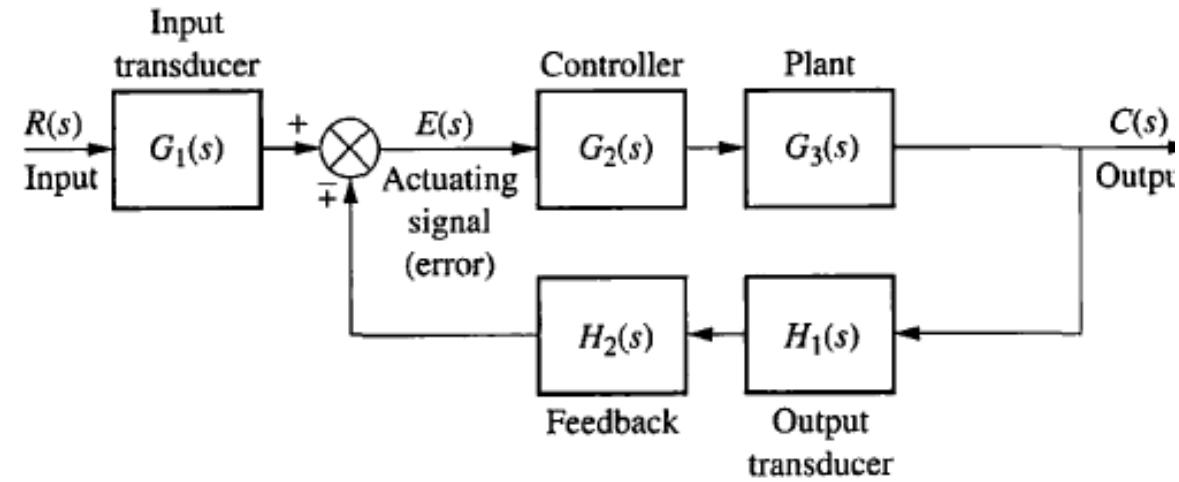
Cascade Form



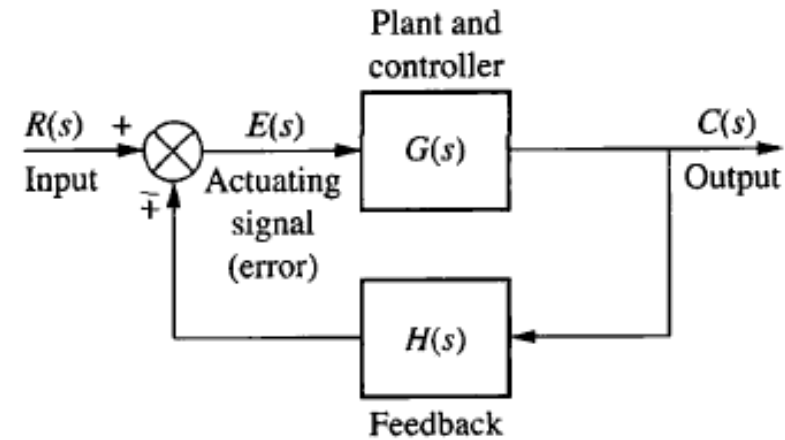
Parallel Form



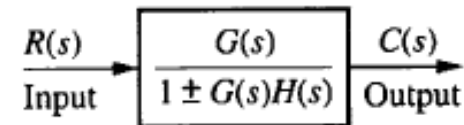
Feedback Form



(a)

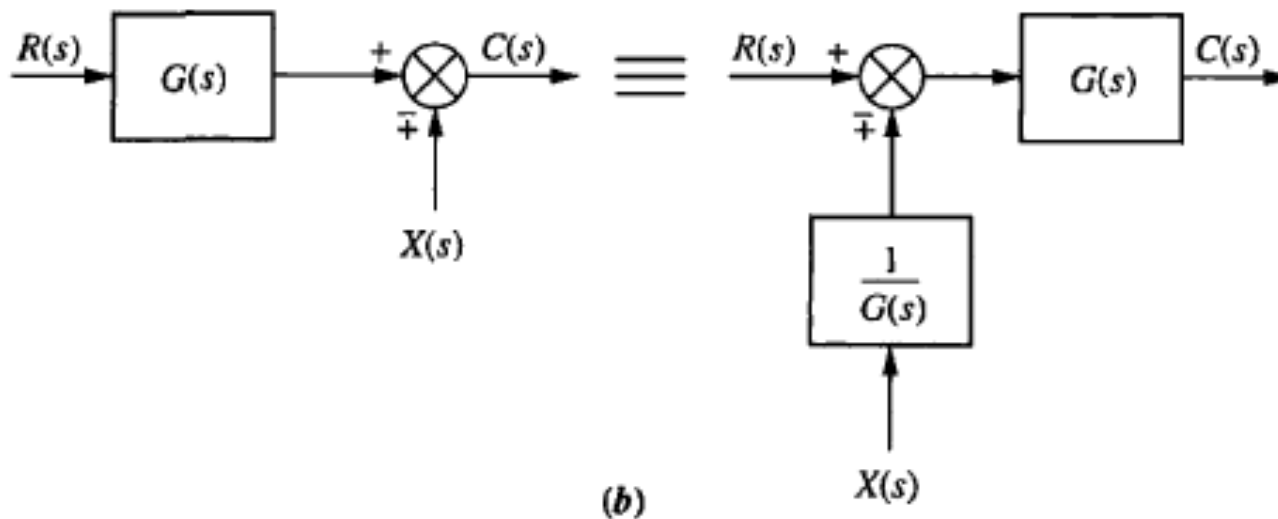
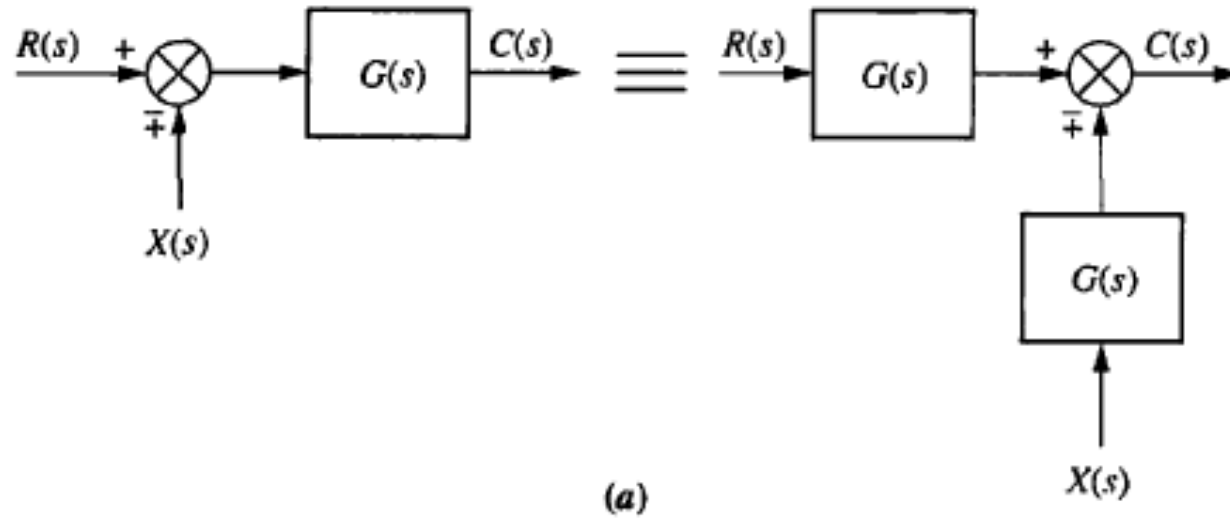


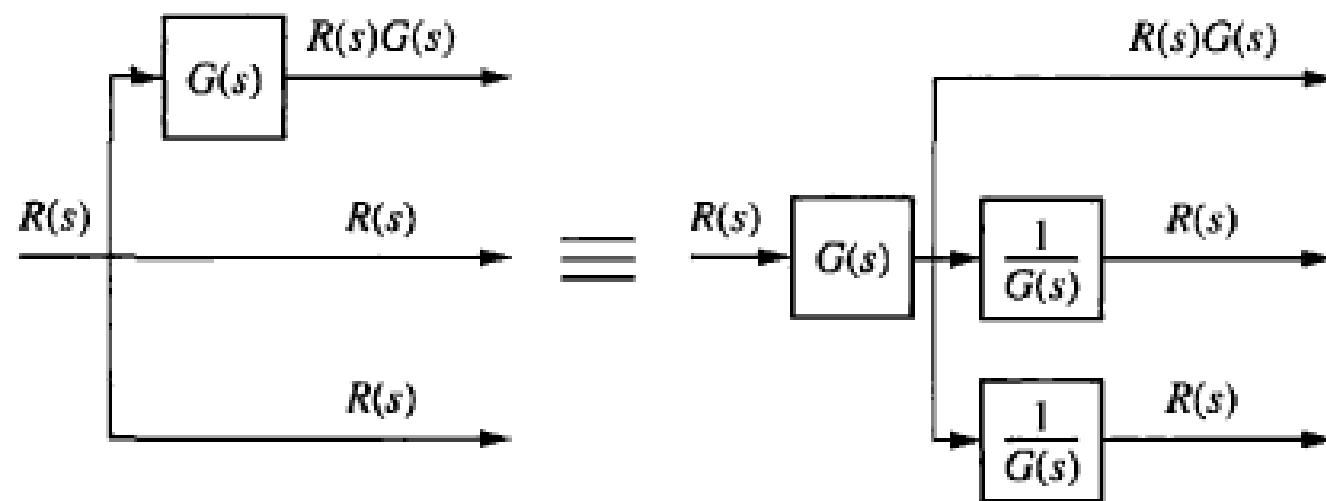
(b)



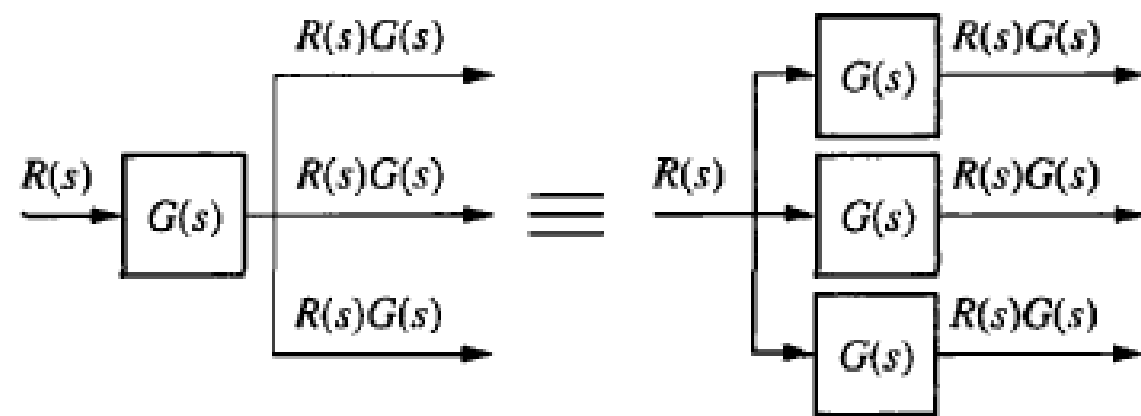
(c)

Bentuk yang Ekuivalen



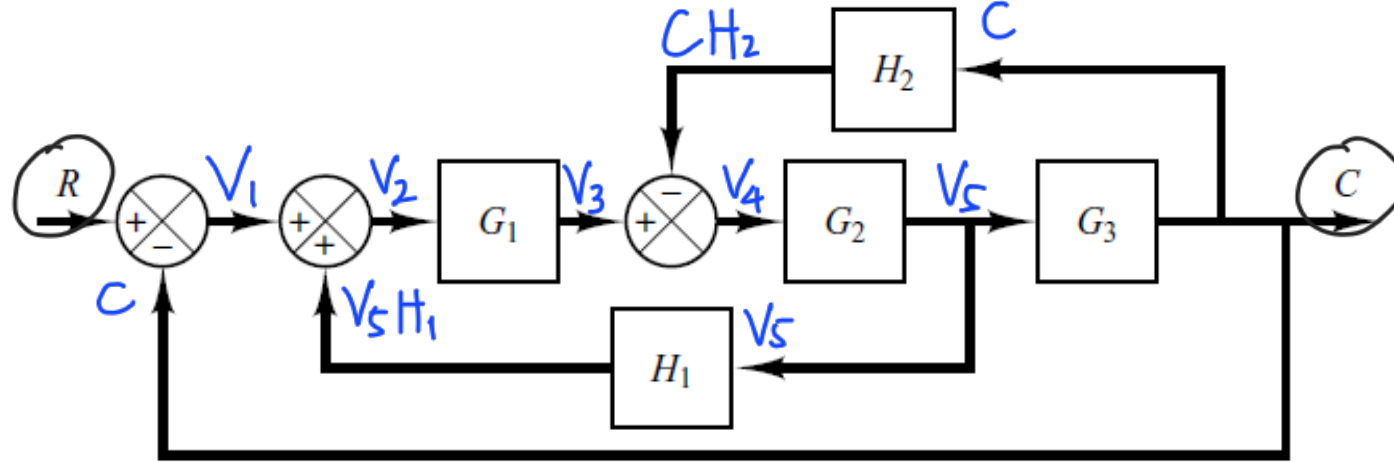


(a)



(b)

Tentukan fungsi transfer dari diagram blok berikut



$$C = V_5 G_3$$

$$\begin{aligned} V_5 &= ((R - C + V_5 H_1) G_1 - C H_2) G_2 \\ &= R G_1 G_2 - C G_1 G_2 + V_5 H_1 G_1 G_2 - C H_2 G_2 \end{aligned}$$

$$\begin{aligned} V_1 &= R - C \quad \text{dan} \quad V_2 = V_1 + V_5 H_1 \\ V_2 &= R - C + V_5 H_1 \end{aligned}$$

$$V_3 = V_2 G_1 = (R - C + V_5 H_1) G_1$$

$$V_4 = V_3 - C H_2 = (R - C + V_5 H_1) G_1 - C H_2$$

$$V_5 = V_4 G_2 = ((R - C + V_5 H_1) G_1 - C H_2) G_2$$

$$V_5 (1 - G_1 G_2 H_1) = R G_1 G_2 - C G_1 G_2 - C H_2 G_2$$

$$V_5 = \frac{G_1 G_2 R - G_1 G_2 C - G_2 H_2 C}{1 - G_1 G_2 H_1}$$

Fungsi transfer: $\frac{C}{R} =$

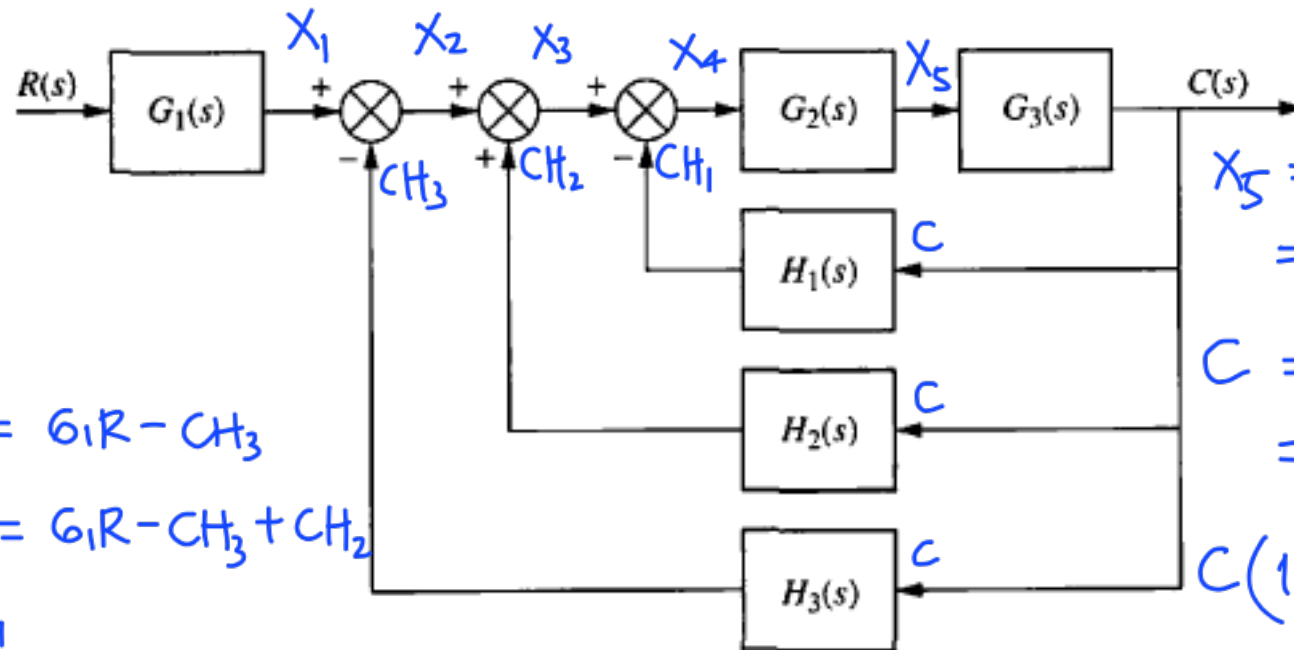
$$C = V_5 G_3 = \frac{G_1 G_2 R - G_1 G_2 C - G_2 H_2 C}{1 - G_1 G_2 H_1} G_3$$

$$C(1 - G_1 G_2 H_1) = G_1 G_2 G_3 R - G_1 G_2 G_3 C - G_2 G_3 H_2 C$$

$$C(1 - G_1 G_2 H_1 + G_1 G_2 G_3 + G_2 G_3 H_2) = G_1 G_2 G_3 R$$

$$\frac{C}{R} = \frac{G_1 G_2 G_3}{1 - G_1 G_2 H_1 + G_1 G_2 G_3 + G_2 G_3 H_2}$$

Latihan



Perhatikan,

$$X_1 = G_1 R$$

$$X_2 = X_1 - CH_3 = G_1 R - CH_3$$

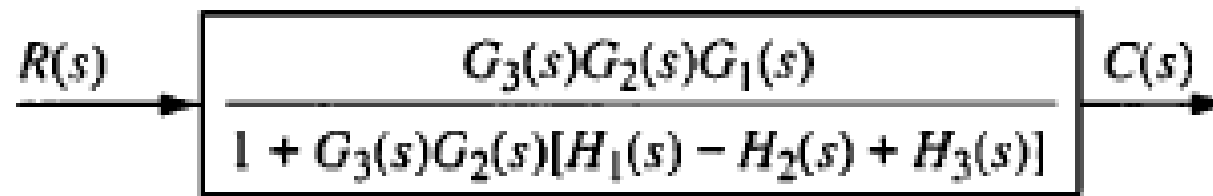
$$X_3 = X_2 + CH_2 = G_1 R - CH_3 + CH_2$$

$$X_4 = X_3 - CH_1 = G_1 R - CH_3 + CH_2 - CH_1$$

$$X_5 = X_4 G_2 = G_2 (G_1 R - CH_3 + CH_2 - CH_1)$$

$$C = X_5 G_3 = G_2 G_3 (G_1 R - CH_3 + CH_2 - CH_1)$$

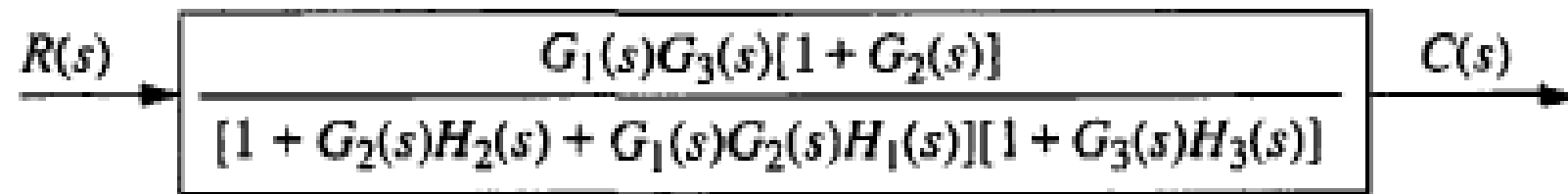
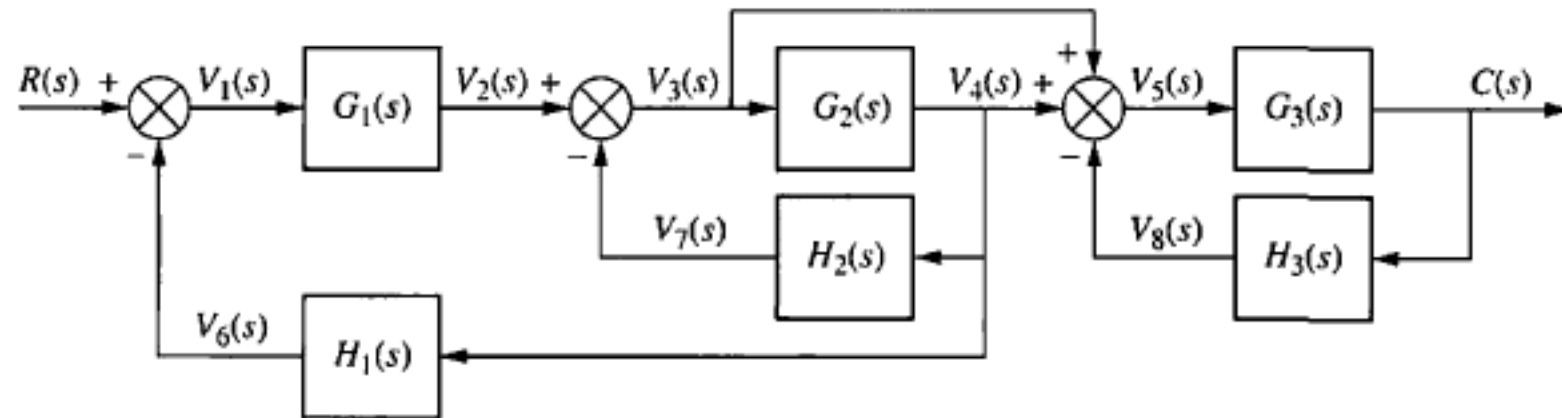
$$C(1 + G_2 G_3 H_1 - G_2 G_3 H_2 + G_2 G_3 H_3) = G_1 G_2 G_3 R$$



(c)

Jadi, $\frac{C}{R} = \frac{G_1 G_2 G_3}{1 + G_2 G_3 (H_1 - H_2 + H_3)}$

Latihan



(e)